# SUPPLEMENTARY MATERIAL FOR "WHERE VIOLENCE PERSISTS: LATENT SPATIAL FIELDS FOR THE RESIDUAL RISK OF ARMED CONFLICT AND SOCIAL UNREST"

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This supplement contains the detailed tables and figures referred to in the main paper. Equation, section and table numbers without the prefix S refer to the main paper.

## S1. Methods.

## S2. Simulation studies.

## S3. Application.

TABLE S1

*Latent blocks of the latent Gaussian model of Section 3.1 of the main paper, their precision matrices and hyperparameters.*

| Block | Dimension | Precision | Hyperparameters |
|---|---|---|---|
| fixed effects $\boldsymbol{\beta}, \delta, \boldsymbol{\gamma}, \boldsymbol{\alpha}$ | $p$ | $10^{-4}\boldsymbol{I}$ | – |
| country effects $\boldsymbol{u}$ | $n_c$ | $\sigma_u^{-2}\boldsymbol{I}$ | $\sigma_u$ |
| spatial field $\boldsymbol{w}$ | $n$ | $\boldsymbol{Q}_s(\rho_w, \sigma_w)$, SPDE precision | $\rho_w, \sigma_w$ |
| cell effects $\boldsymbol{e}$ | $n$ | $\sigma_e^{-2}\boldsymbol{I}$ | $\sigma_e$ |
| SVC fields $\boldsymbol{\eta}_k$ | $n$ each | $\boldsymbol{Q}_s(\rho_k, \sigma_k)$ | $\rho_k, \sigma_k$ |
| space-time field $\boldsymbol{v}$ | $n\,T_v$ | $\boldsymbol{Q}_{\mathrm{AR}}(\phi) \otimes \boldsymbol{Q}_s(\rho_v, \sigma_v)$ | $\rho_v, \sigma_v, \phi$ |

TABLE S2

*Boundary treatments on a* $60 \times 30$ *lattice of* $0.5^\circ$ *cells with a sea strait of 2, 4 or 6 cells (about 110, 220, 330 km): marginal variance of a coastal and of an inland land cell (nominal 1) and correlation between the two coastal cells facing each other across the strait, from 4,000 exact draws with* $\rho = 400$ *km. "Land only" removes the strait from the lattice; "sea = land" keeps it as ordinary cells without data. The strait lies inside a closed rectangle. This is not the geometry of the application, which has an open coast, a buffer of sea cells and a reflecting outer edge, and the fitted ranges there are both above and below 400 km, so the table shows the behaviour of the barrier only qualitatively.*

| Strait | | Land only | Sea = land | $f = 0.1$ | $f = 0.2$ | $f = 0.5$ |
|---|---|---|---|---|---|---|
| 2 cells | var. coast | 1.98 | 1.07 | 1.80 | 1.67 | 1.29 |
| | var. inland | 1.10 | 1.06 | 1.09 | 1.11 | 1.09 |
| | corr. across | -0.01 | 0.49 | -0.03 | 0.01 | 0.16 |
| 4 cells | var. coast | 2.00 | 1.08 | 1.77 | 1.69 | 1.29 |
| | var. inland | 1.06 | 1.05 | 1.09 | 1.09 | 1.09 |
| | corr. across | 0.00 | 0.28 | 0.01 | 0.01 | 0.04 |
| 6 cells | var. coast | 2.02 | 1.04 | 1.76 | 1.66 | 1.26 |
| | var. inland | 1.06 | 1.05 | 1.08 | 1.07 | 1.07 |
| | corr. across | -0.02 | 0.14 | 0.01 | 0.01 | 0.01 |

TABLE S3

*S1: residual diagnostics and recovery of the field (mean and SD over 20 replicates).*

| Quantity | Mean | SD |
|---|---|---|
| Moran's $I$ of cell residuals, XGBoost as in the original | 0.360 | 0.050 |
| Moran's $I$ of cell residuals, cross-fitted XGBoost | 0.637 | 0.067 |
| $\mathrm{cor}(\hat{w}, w)$, hybrid with $u + w$ | 0.849 | 0.044 |
| $\hat{\rho}_w$ (km) [truth 400] | 409 | 50 |
| $\hat{\sigma}_w$ [truth 1] | 0.829 | 0.069 |

TABLE S4

*S2: estimation of the effect $\alpha$ of a national indicator (30 replicates per scenario). Coverage of nominal 95% intervals, and the proportion of replicates in which the interval excludes zero. $^*$The model with $u + w$, in which $Z_0$ is replaced by its residual after projection on the 10 smoothest eigenvectors of the lattice Laplacian. As in the application, $u$ stays in the model, so the coefficient is identified partly through the shrinkage of $u$ under its exchangeable prior; we read spatial+ as a sensitivity analysis. $^\dagger$Least-squares slope of SHAP values on $Z_0$, with the 95% block-bootstrap percentile interval and interquartile range.*

| Estimator | Mean | Emp. SD | Mean SE | Coverage | Reject $\alpha = 0$ |
|---|---|---|---|---|---|
| *No confounding,* $\alpha = 0$ | | | | | |
| Logistic regression | -0.007 | 0.107 | 0.016 | 0.23 | 0.77 |
| + country effects $u$ | 0.016 | 0.109 | 0.123 | 0.97 | 0.03 |
| + spatial field $w$ | 0.002 | 0.065 | 0.052 | 0.93 | 0.07 |
| + $u$ + $w$ | 0.000 | 0.063 | 0.081 | 1.00 | 0.00 |
| spatial+: $u + w$, $Z_0$ projected$^*$ | 0.002 | 0.072 | 0.082 | 0.97 | 0.03 |
| XGBoost SHAP slope$^\dagger$ | -0.006 | 0.063 | 0.044 | 0.80 | 0.20 |
| *Confounding,* $\alpha = 0$ | | | | | |
| Logistic regression | 0.465 | 0.115 | 0.018 | 0.00 | 1.00 |
| + country effects $u$ | 0.493 | 0.098 | 0.078 | 0.00 | 1.00 |
| + spatial field $w$ | 0.200 | 0.098 | 0.056 | 0.30 | 0.70 |
| + $u$ + $w$ | 0.200 | 0.095 | 0.077 | 0.33 | 0.67 |
| spatial+: $u + w$, $Z_0$ projected$^*$ | 0.153 | 0.092 | 0.079 | 0.60 | 0.40 |
| XGBoost SHAP slope$^\dagger$ | 0.332 | 0.132 | 0.056 | 0.00 | 1.00 |
| *Confounding,* $\alpha = -0.3$ | | | | | |
| Logistic regression | 0.219 | 0.114 | 0.017 | 0.00 | 1.00 |
| + country effects $u$ | 0.235 | 0.104 | 0.079 | 0.00 | 0.80 |
| + spatial field $w$ | -0.090 | 0.100 | 0.056 | 0.20 | 0.40 |
| + $u$ + $w$ | -0.091 | 0.098 | 0.077 | 0.37 | 0.33 |
| spatial+: $u + w$, $Z_0$ projected$^*$ | -0.130 | 0.095 | 0.078 | 0.50 | 0.40 |
| XGBoost SHAP slope$^\dagger$ | 0.138 | 0.091 | 0.049 | 0.00 | 0.87 |

## S4. Additional results.

## S5. Further simulation results.

TABLE S5

*S4: estimation of biome-specific temperature effects (30 replicates). "Detects sign": proportion of replicates in which the interval excludes zero on the correct side. ‡Median SHAP value of raw temperature within the biome, with the interquartile range of the bootstrap replicates, as in the original analysis; it estimates neither $\gamma_b$ nor $\bar{\beta}_b$.*

| Estimator / biome | Mean est. | Cov. $\gamma_b$ | Cov. $\bar{\beta}_b$ | Detects sign |
|---|---|---|---|---|
| *Biome slopes, no latent effects* | | | | |
| biome 1 ($\gamma = -0.3$) | -0.273 | 0.57 | 0.77 | 1.00 |
| biome 2 ($\gamma = -0.1$) | -0.083 | 0.57 | 0.83 | 0.60 |
| biome 3 ($\gamma = +0.1$) | 0.108 | 0.53 | 0.70 | 0.73 |
| biome 4 ($\gamma = +0.3$) | 0.307 | 0.50 | 0.53 | 1.00 |
| *Biome slopes + $u$ + $w$* | | | | |
| biome 1 ($\gamma = -0.3$) | -0.286 | 0.53 | 0.87 | 1.00 |
| biome 2 ($\gamma = -0.1$) | -0.082 | 0.57 | 0.90 | 0.67 |
| biome 3 ($\gamma = +0.1$) | 0.119 | 0.60 | 0.73 | 0.73 |
| biome 4 ($\gamma = +0.3$) | 0.319 | 0.53 | 0.87 | 1.00 |
| *SVC: biome means + $\eta$ + $u$ + $w$* | | | | |
| biome 1 ($\gamma = -0.3$) | -0.277 | 0.97 | 1.00 | 0.93 |
| biome 2 ($\gamma = -0.1$) | -0.077 | 0.93 | 1.00 | 0.23 |
| biome 3 ($\gamma = +0.1$) | 0.111 | 0.90 | 1.00 | 0.33 |
| biome 4 ($\gamma = +0.3$) | 0.309 | 0.83 | 1.00 | 0.97 |
| *XGBoost SHAP slope (targets $\gamma_b/2$)* | | | | |
| biome 1 ($\gamma = -0.3$) | -0.092 | 0.03 | 0.03 | 0.60 |
| biome 2 ($\gamma = -0.1$) | -0.026 | 0.47 | 0.53 | 0.23 |
| biome 3 ($\gamma = +0.1$) | 0.048 | 0.47 | 0.40 | 0.37 |
| biome 4 ($\gamma = +0.3$) | 0.091 | 0.03 | 0.00 | 0.60 |
| *XGBoost SHAP slope, interactions credited* | | | | |
| biome 1 ($\gamma = -0.3$) | -0.191 | 0.27 | 0.40 | 0.83 |
| biome 2 ($\gamma = -0.1$) | -0.056 | 0.63 | 0.63 | 0.30 |
| biome 3 ($\gamma = +0.1$) | 0.089 | 0.57 | 0.73 | 0.53 |
| biome 4 ($\gamma = +0.3$) | 0.176 | 0.40 | 0.30 | 0.80 |
| *XGBoost median SHAP, raw temperature*‡ | | | | |
| biome 1 ($\gamma = -0.3$) | -0.052 | 0.03 | 0.07 | 0.57 |
| biome 2 ($\gamma = -0.1$) | -0.151 | 0.13 | 0.13 | 0.97 |
| biome 3 ($\gamma = +0.1$) | -0.150 | 0.00 | 0.00 | 0.00 |
| biome 4 ($\gamma = +0.3$) | -0.043 | 0.00 | 0.00 | 0.20 |

TABLE S6

*S4: recovery of the spatially varying coefficient $\beta(s)$ (mean and SD over 30 replicates).*

| Quantity | Mean | SD |
|---|---|---|
| $\mathrm{cor}\{\hat{\beta}(s), \beta(s)\}$, SVC | 0.918 | 0.036 |
| RMSE of $\hat{\beta}(s)$, SVC | 0.106 | 0.010 |
| RMSE of $\hat{\beta}(s)$, biome slopes only | 0.147 | 0.020 |
| $\hat{\sigma}_\eta$ [truth 0.15] | 0.133 | 0.025 |
| $\hat{\rho}_\eta$ (km) [truth 600] | 564 | 261 |

TABLE S7

*Test-period (2023–2024) residual diagnostics, AME region, for every model on identical test rows: Moran's $I$ of the standardised cell residuals (permutation p-value from 999 permutations, bounded below by 0.001) and the residual correlation between neighbouring cells in the same country and across a border.*

| Outcome | Model | Moran's $I$ | $p$ | Neighbour residual corr. | |
|---|---|---|---|---|---|
| | | | | within country | across border |
| Conflict | XGBoost, corrected | 0.424 | ≤0.001 | 0.449 | 0.214 |
| Conflict | XGBoost + cell event rate | 0.295 | ≤0.001 | 0.301 | 0.229 |
| Conflict | XGBoost, cross-fitted by country | 0.539 | ≤0.001 | 0.585 | 0.285 |
| Conflict | Cross-fitted XGBoost + $u$ + iid cell effects | 0.450 | ≤0.001 | 0.476 | 0.115 |
| Conflict | Cross-fitted XGBoost + $u + w$ | 0.449 | ≤0.001 | 0.470 | 0.136 |
| Conflict | xgb_dedup | 0.354 | ≤0.001 | 0.372 | 0.226 |
| Unrest | XGBoost, corrected | 0.148 | ≤0.001 | 0.161 | 0.081 |
| Unrest | XGBoost + cell event rate | 0.253 | ≤0.001 | 0.277 | 0.106 |
| Unrest | XGBoost, cross-fitted by country | 0.258 | ≤0.001 | 0.287 | 0.097 |
| Unrest | Cross-fitted XGBoost + $u$ + iid cell effects | 0.292 | ≤0.001 | 0.333 | -0.013 |
| Unrest | Cross-fitted XGBoost + $u + w$ | 0.287 | ≤0.001 | 0.327 | -0.016 |
| Unrest | xgb_dedup | 0.156 | ≤0.001 | 0.166 | 0.117 |

TABLE S8

*Fitted hyperparameters of the hybrid models, AME region: range $\rho_w$ and standard deviation $\sigma_w$ of the field, standard deviation of the independent cell effects, and standard deviation $\sigma_u$ of the country effects.*

| Outcome | Latent terms | $\hat{\rho}_w$ (km) | $\hat{\sigma}_w$ | $\hat{\sigma}_{\text{cell}}$ | $\hat{\sigma}_u$ |
|---|---|---|---|---|---|
| Conflict | $u + w$ | 263 | 1.48 | – | 0.76 |
| Conflict | $u$ + iid cell effects | – | – | 1.30 | 1.37 |
| Unrest | $u + w$ | 65 | 1.25 | – | 1.01 |
| Unrest | $u$ + iid cell effects | – | – | 1.15 | 1.06 |

TABLE S9

*Posterior summaries of the field $w$ of the hybrid model averaged over regions: mean, standard deviation and posterior probability that the regional average is positive. The regions were chosen before the fit, as known conflict zones and regions expected to be quiet. The last two rows give the share of all cells whose posterior probability of $w(s) > 0$ is above 0.95 or below 0.05.*

| Region | Longitude | Latitude | Cells | Conflict | | Unrest | |
|---|---|---|---|---|---|---|---|
| | | | | mean (SD) | $P(>0)$ | mean (SD) | $P(>0)$ |
| Eastern DRC | 27 to 30.5 | -5 to 2 | 94 | 1.49 (0.18) | 1.00 | 0.52 (0.11) | 1.00 |
| Northern Mozambique | 38 to 41 | -14 to -10 | 39 | 1.44 (0.29) | 1.00 | 0.14 (0.20) | 0.76 |
| Syria and Iraq | 36 to 46 | 32 to 38 | 240 | 2.41 (0.21) | 1.00 | -0.03 (0.10) | 0.36 |
| Yemen | 43 to 49 | 13 to 17 | 84 | 1.95 (0.37) | 1.00 | 0.04 (0.16) | 0.61 |
| Lake Chad | 12 to 16 | 10 to 14 | 64 | 1.66 (0.19) | 1.00 | -0.19 (0.13) | 0.07 |
| Central Sahel | -5 to 5 | 11 to 17 | 240 | 1.56 (0.18) | 1.00 | -0.03 (0.09) | 0.39 |
| Horn of Africa | 38 to 48 | 3 to 10 | 277 | 0.41 (0.21) | 0.97 | -0.04 (0.09) | 0.32 |
| Sudan | 23 to 35 | 10 to 16 | 288 | -0.06 (0.23) | 0.40 | 0.09 (0.09) | 0.83 |
| Arabian interior | 42 to 55 | 18 to 27 | 423 | -0.61 (0.30) | 0.02 | -0.03 (0.09) | 0.38 |
| Western Congo basin | 14 to 24 | -5 to 3 | 320 | -1.10 (0.22) | 0.00 | -0.13 (0.09) | 0.07 |
| Southern Africa | 18 to 32 | -34 to -22 | 619 | -0.45 (0.26) | 0.04 | 0.00 (0.08) | 0.53 |
| Share of cells with $P\{w(s) > 0\} > 0.95$ | | | 13719 | 0.189 | | 0.116 | |
| Share of cells with $P\{w(s) > 0\} < 0.05$ | | | 13719 | 0.030 | | 0.026 | |

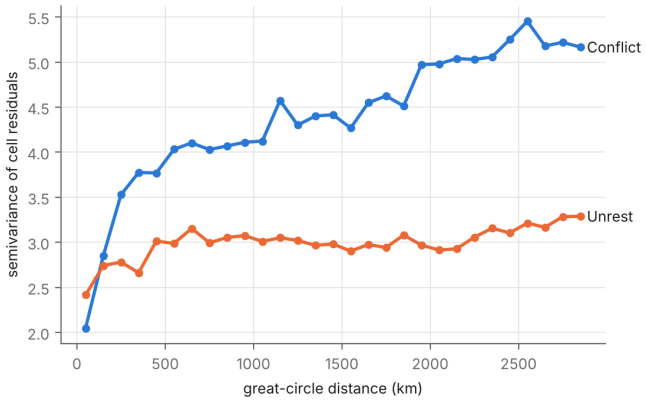



FIG S1. *Empirical semivariograms of the standardised test-period cell residuals, against great-circle distance.*

TABLE S10

*Dynamic experiments for the hybrid models, AME region. In all of them the offset is the country cross-fitted XGBoost prediction. (a) Oracle: the latent terms are fitted to the 2023–2024 test weeks themselves, with the offset fixed, to show how much residual structure the test period contains and whether it is smooth in space; the correlation compares this field with the field fitted on 2018–2022, and Moran's $I$ is computed on the in-sample cell residuals. (b) Scores on the 2024 test weeks only (713,388 rows per outcome), for models fitted on 2018–2022 and for models whose offset and latent terms are retrained on 2018–2023 (training years in parentheses); differences in log score are relative to the corrected XGBoost model, with SEs clustered by country. (c, d) Hyperparameters of the space-time hybrids: offset, $u$, independent cell effects $e$ and an AR(1) $\otimes$ Matérn field $v$ on $1^\circ$ cells, with no static field $w$. The field $v$ is defined on the observed cells only, with reflecting edges, which departs from the boundary rule of Remark 1 of the main paper; $\phi$ is the AR(1) coefficient per time step, and the number of steps in parentheses is the number that falls in the training period. (e) Quarterly update: the latent terms are re-estimated at the start of each quarter of the test period with all data observed so far, and the hyperparameters are kept at their 2018–2022 values. Out-of-sample scores of the models in (c), (d) and (e) are in the forecasting comparison of Section 5 of the main paper.*

| Quantity | Conflict | Unrest |
|---|---|---|
| *(a) Oracle: latent terms fitted to the 2023–2024 test weeks, offset fixed* | | |
| log marginal likelihood, $u + w$ minus $u$ + iid cell effects | 1045.4 | 69.0 |
| $\hat{\rho}_w$ (km) | 374 | 66 |
| $\hat{\sigma}_w$ | 1.49 | 1.08 |
| cor(oracle $\hat{w}$, $\hat{w}$ fitted on 2018–2022) | 0.79 | 0.69 |
| Moran's $I$ of cell residuals, offset alone | 0.539 | 0.258 |
| Moran's $I$ of cell residuals, $u$ + iid cell effects | 0.346 | 0.154 |
| Moran's $I$ of cell residuals, $u + w$ | $-0.119$ | 0.003 |
| *(b) 2024 test weeks only: AUC-PR; $\Delta$ log score $\times 10^3$ (SE)* | | |
| XGBoost, corrected (2018–2022) | 0.522; reference | 0.553; reference |
| XGBoost + cell event rate (2018–2022) | 0.533; $-2.78$ (1.42) | 0.562; $-2.34$ (0.96) |
| Offset + $u + w$ (2018–2022) | 0.520; $-0.67$ (1.22) | 0.557; $-1.73$ (0.67) |
| Cross-fitted XGBoost (2018–2023) | 0.487; 6.27 (1.46) | 0.526; 3.59 (1.11) |
| Offset + $u$ + iid cell effects (2018–2023) | 0.535; $-1.85$ (0.76) | 0.572; $-3.24$ (0.79) |
| Offset + $u + w$ (2018–2023) | 0.535; $-2.17$ (0.78) | 0.572; $-3.27$ (0.79) |
| $\hat{\rho}_w$ (km), $u + w$ (2018–2023) | 248 | 66 |
| Test rows (2024) | 713,388 | 713,388 |
| *(c) Space-time hybrid, offset + $u$ + cell effects + $v$, 13-week steps on observed $1^\circ$ cells* | | |
| $\hat{\rho}_v$ (km) | 779 | 996 |
| $\hat{\sigma}_v$ | 1.39 | 0.62 |
| $\hat{\phi}$ (per 13 weeks) | 0.986 | 0.949 |
| $\hat{\sigma}_e$ | 0.81 | 1.07 |
| $\hat{\sigma}_u$ | 0.79 | 0.96 |
| Time steps (training) | 29 (21) | 29 (21) |
| Fit time (h) | 2.2 | 2.3 |
| *(d) Space-time hybrid, offset + $u$ + cell effects + $v$, 26-week steps on observed $1^\circ$ cells* | | |
| $\hat{\rho}_v$ (km) | 731 | 891 |
| $\hat{\sigma}_v$ | 1.33 | 0.63 |
| $\hat{\phi}$ (per 26 weeks) | 0.974 | 0.943 |
| $\hat{\sigma}_e$ | 0.80 | 1.06 |
| $\hat{\sigma}_u$ | 0.79 | 0.94 |
| Time steps (training) | 15 (11) | 15 (11) |
| Fit time (h) | 0.4 | 0.3 |
| *(e) Quarterly update: hyperparameters fixed at their 2018–2022 values* | | |
| $\hat{\rho}_w$ (km) | 263 | 65 |
| $\hat{\sigma}_w$ | 1.48 | 1.25 |
| Number of updates | 8 | 8 |

TABLE S11

*S5: full results for the escalation effect $\kappa$ of recent unrest on conflict, by scenario and estimator, over 20 replicates per scenario. Mean $\hat{\kappa}$, its empirical SD across replicates and the mean of the estimated SEs; coverage of the nominal 95% intervals for $\kappa$ with its Monte Carlo SE; proportion of replicates whose interval excludes zero; mean estimate of the own-lag coefficient $d_C$ (truth 0.3); mean and SD of the estimated loading $\lambda$ of the shared space-time field (truth 0.8, joint model only); and mean log score, AUC-PR and onset AUC-PR of the conflict forecasts over the 26 test weeks. Onset rows are those of cells without a conflict event in the previous 52 weeks. The last two scenarios add unrest-only episodes (AR(1) with coefficient 0.95, Matérn innovations of range 200 km, marginal SD 0.6), which the joint model with one shared field must absorb into that field.*

| Estimator | Mean $\hat{\kappa}$ | Emp. SD | Mean SE | Coverage (MC SE) | Reject $\kappa = 0$ | Mean $\hat{d}_c$ | Mean $\hat{\lambda}$ (SD) | Log score | AUC-PR | Onset AUC-PR |
|---|---|---|---|---|---|---|---|---|---|---|
| *No escalation, $\kappa = 0$* | | | | | | | | | | |
| Logistic regression | 0.695 | 0.193 | 0.129 | 0.05 (0.05) | 0.95 | 1.54 | – | 0.1372 | 0.063 | 0.022 |
| Static field $w_c$ | 0.426 | 0.161 | 0.135 | 0.15 (0.08) | 0.85 | 0.82 | – | 0.1335 | 0.090 | 0.037 |
| Static field $w_c$ + space-time field | 0.269 | 0.158 | 0.140 | 0.45 (0.11) | 0.55 | 0.38 | – | 0.1341 | 0.090 | 0.034 |
| Joint model, shared space-time field | 0.042 | 0.166 | 0.138 | 0.90 (0.07) | 0.10 | 0.38 | 0.83 (0.14) | 0.1343 | 0.088 | 0.033 |
| Joint model, shared + unrest-specific space-time fields | 0.058 | 0.165 | 0.139 | 0.85 (0.08) | 0.15 | 0.36 | 0.77 (0.14) | 0.1342 | 0.089 | 0.035 |
| *Escalation, $\kappa = 0.4$* | | | | | | | | | | |
| Logistic regression | 1.026 | 0.163 | 0.116 | 0.00 (0.00) | 1.00 | 1.52 | – | 0.1430 | 0.077 | 0.025 |
| Static field $w_c$ | 0.794 | 0.128 | 0.122 | 0.10 (0.07) | 1.00 | 0.80 | – | 0.1392 | 0.105 | 0.044 |
| Static field $w_c$ + space-time field | 0.659 | 0.108 | 0.127 | 0.50 (0.11) | 1.00 | 0.37 | – | 0.1399 | 0.105 | 0.045 |
| Joint model, shared space-time field | 0.434 | 0.111 | 0.126 | 0.95 (0.05) | 0.95 | 0.36 | 0.83 (0.15) | 0.1401 | 0.103 | 0.042 |
| Joint model, shared + unrest-specific space-time fields | 0.450 | 0.109 | 0.126 | 0.95 (0.05) | 0.95 | 0.35 | 0.78 (0.14) | 0.1400 | 0.104 | 0.044 |
| *No escalation, $\kappa = 0$, with unrest-only episodes* | | | | | | | | | | |
| Logistic regression | 0.683 | 0.201 | 0.121 | 0.10 (0.07) | 0.90 | 1.53 | – | 0.1373 | 0.063 | 0.025 |
| Static field $w_c$ | 0.411 | 0.164 | 0.128 | 0.20 (0.09) | 0.80 | 0.82 | – | 0.1335 | 0.089 | 0.037 |
| Static field $w_c$ + space-time field | 0.281 | 0.158 | 0.135 | 0.45 (0.11) | 0.55 | 0.38 | – | 0.1342 | 0.090 | 0.035 |
| Joint model, shared space-time field | $-0.012$ | 0.166 | 0.131 | 0.85 (0.08) | 0.15 | 0.47 | 1.19 (0.32) | 0.1343 | 0.088 | 0.038 |
| Joint model, shared + unrest-specific space-time fields | 0.064 | 0.169 | 0.133 | 0.85 (0.08) | 0.15 | 0.37 | 0.81 (0.20) | 0.1344 | 0.089 | 0.036 |
| *Escalation, $\kappa = 0.4$, with unrest-only episodes* | | | | | | | | | | |
| Logistic regression | 1.017 | 0.175 | 0.108 | 0.00 (0.00) | 1.00 | 1.51 | – | 0.1435 | 0.078 | 0.031 |
| Static field $w_c$ | 0.786 | 0.140 | 0.115 | 0.15 (0.08) | 1.00 | 0.80 | – | 0.1396 | 0.107 | 0.048 |
| Static field $w_c$ + space-time field | 0.679 | 0.136 | 0.122 | 0.35 (0.11) | 1.00 | 0.37 | – | 0.1403 | 0.107 | 0.046 |
| Joint model, shared space-time field | 0.385 | 0.125 | 0.118 | 0.90 (0.07) | 0.90 | 0.44 | 1.17 (0.29) | 0.1406 | 0.105 | 0.043 |
| Joint model, shared + unrest-specific space-time fields | 0.467 | 0.137 | 0.120 | 0.85 (0.08) | 1.00 | 0.35 | 0.81 (0.19) | 0.1406 | 0.106 | 0.045 |

TABLE S12

*Joint model of conflict and unrest, by region (Section 5 of the main paper): all hyperparameters and lag coefficients (posterior mean, with SE in parentheses) of the two univariate conflict models and of the joint model, fitted on 2018–2022. Subscripts C and U denote conflict and unrest. In the univariate model with $u + e + w + v$, the space-time field $v$ belongs to conflict alone; in the joint model it is shared, with loading $\lambda$ for unrest. Histories are $\log(1 +$ events in the previous four weeks$)$; neighbour histories are means over the rook neighbours, standardised on the training rows. The log marginal likelihood of the joint model includes the unrest data and is not comparable with those of the univariate models. Test scores are for 2023–2024 in the same region. The field $v$ is defined on the observed $1^\circ$ cells only, with reflecting edges, which departs from the boundary rule of Remark 1 of the main paper. SEs are plug-in values; in S3 and S5 such SEs are too small when a space-time field is fitted (ratio of empirical SD to mean SE up to 1.66 in S3 and 1.20 in S5 without escalation), so intervals from these models are likely too narrow. With a single shared space-time field, unrest-only episodes are absorbed into $v$, which can bias $\kappa$ towards zero; in S5 they inflated $\hat{\lambda}$ instead. [TODO: biv2 results.]*

| Quantity | Conflict only $u + e + w$ | Conflict only $u + e + w + v$ | Joint shared $v$ |
|---|---|---|---|
| *West Africa and Sahel* | | | |
| 2,505 cells, 22 countries, 906,810 rows per outcome | | | |
| $v$ on 641 observed $1^\circ$ cells (reflecting edges), 28 steps of 13 weeks | | | |
| Hyperparameters | | | |
| $\sigma_{u,\mathrm{C}}$ | 0.69 | 0.67 | 0.69 |
| $\sigma_{e,\mathrm{C}}$ | 0.65 | 0.64 | 0.66 |
| $\rho_{w,\mathrm{C}}$ (km) | 514 | 508 | 453 |
| $\sigma_{w,\mathrm{C}}$ | 1.70 | 1.74 | 1.39 |
| $\sigma_{u,\mathrm{U}}$ | – | – | 0.53 |
| $\sigma_{e,\mathrm{U}}$ | – | – | 1.34 |
| $\rho_{w,\mathrm{U}}$ (km) | – | – | 369 |
| $\sigma_{w,\mathrm{U}}$ | – | – | 0.21 |
| $\rho_v$ (km) | – | 423 | 499 |
| $\sigma_v$ | – | 0.69 | 1.01 |
| $\phi$ (per step) | – | 0.905 | 0.952 |
| $\lambda$ | – | – | 0.868 |
| Conflict equation | | | |
| $\delta_{\mathrm{C}}$: own history | 0.575 (0.016) | 0.444 (0.017) | 0.459 (0.017) |
| $\kappa$: unrest history | 0.138 (0.031) | 0.131 (0.032) | 0.044 (0.032) |
| $\delta_{\mathrm{C}}^{\mathrm{nb}}$: neighbours' conflict | 0.129 (0.007) | 0.051 (0.008) | 0.061 (0.008) |
| $\kappa^{\mathrm{nb}}$: neighbours' unrest | 0.005 (0.008) | 0.003 (0.008) | −0.020 (0.008) |
| Unrest equation | | | |
| $\delta_{\mathrm{U}}$: own history | – | – | 0.351 (0.029) |
| $\kappa_{\mathrm{U}}$: conflict history | – | – | 0.066 (0.031) |
| $\delta_{\mathrm{U}}^{\mathrm{nb}}$: neighbours' unrest | – | – | 0.017 (0.009) |
| $\kappa_{\mathrm{U}}^{\mathrm{nb}}$: neighbours' conflict | – | – | −0.042 (0.013) |
| Fit | | | |
| Log marginal likelihood | −49530.1 | −49080.6 | −76591.8 |
| Fit time (h) | 0.1 | 0.7 | 4.5 |
| Test AUC-PR, conflict | 0.470 | 0.471 | 0.466 |
| Test log score, conflict | 0.1020 | 0.1026 | 0.1031 |
| Test AUC-PR, unrest | – | – | 0.289 |
| Test log score, unrest | – | – | 0.0490 |
| Joint model with an unrest-specific space-time field: [TODO: biv2 results] | | | |
| *AME region*: [TODO: running] | | | |

TABLE S13

*S3: estimation of the state-dependence parameter $\delta$ for all five truths and all step lengths of the space-time field $v$ (20 replicates per truth). The two truths with slow $v$ have an AR(1) coefficient of 0.97 per week instead of 0.9. SD/SE is the ratio of the empirical SD of $\hat{\delta}$ to the mean estimated SE; values above 1 mean that the plug-in intervals are too narrow. Time is the mean fit time per replicate.*

| Estimator | Mean $\hat{\delta}$ | Emp. SD | Mean SE | SD/SE | Coverage | Time (s) |
|---|---|---|---|---|---|---|
| *State dependence ($\delta = 1$)* | | | | | | |
| Lags only | 1.263 | 0.107 | 0.136 | 0.78 | 0.60 | 0.1 |
| Lags + $w$ | 1.032 | 0.131 | 0.139 | 0.94 | 0.90 | 0.9 |
| Lags + $w$ + $v$ (8-week steps) | 0.981 | 0.128 | 0.140 | 0.92 | 0.95 | 27.3 |
| Lags + $w$ + $v$ (4-week steps) | 0.990 | 0.135 | 0.140 | 0.96 | 0.95 | 101.5 |
| Lags + $w$ + $v$ (weekly steps) | 0.997 | 0.143 | 0.140 | 1.02 | 0.95 | 449.6 |
| *Heterogeneity ($\delta = 0$)* | | | | | | |
| Lags only | 1.614 | 0.386 | 0.099 | 3.88 | 0.00 | 0.1 |
| Lags + $w$ | 0.464 | 0.093 | 0.111 | 0.84 | 0.00 | 1.0 |
| Lags + $w$ + $v$ (8-week steps) | 0.097 | 0.140 | 0.117 | 1.19 | 0.80 | 26.4 |
| Lags + $w$ + $v$ (4-week steps) | 0.045 | 0.152 | 0.119 | 1.28 | 0.90 | 95.1 |
| Lags + $w$ + $v$ (weekly steps) | 0.011 | 0.162 | 0.120 | 1.35 | 0.90 | 474.2 |
| *Mixed ($\delta = 0.5$)* | | | | | | |
| Lags only | 1.659 | 0.351 | 0.106 | 3.32 | 0.00 | 0.1 |
| Lags + $w$ | 0.782 | 0.111 | 0.115 | 0.96 | 0.25 | 1.0 |
| Lags + $w$ + $v$ (8-week steps) | 0.567 | 0.144 | 0.119 | 1.21 | 0.85 | 24.6 |
| Lags + $w$ + $v$ (4-week steps) | 0.545 | 0.152 | 0.120 | 1.27 | 0.85 | 90.0 |
| Lags + $w$ + $v$ (weekly steps) | 0.535 | 0.172 | 0.120 | 1.43 | 0.80 | 349.7 |
| *Heterogeneity, slow $v$ ($\delta = 0$, $\phi = 0.97$ per week)* | | | | | | |
| Lags only | 1.649 | 0.418 | 0.100 | 4.18 | 0.00 | 0.1 |
| Lags + $w$ | 0.384 | 0.115 | 0.113 | 1.01 | 0.05 | 1.0 |
| Lags + $w$ + $v$ (8-week steps) | 0.037 | 0.160 | 0.119 | 1.35 | 0.90 | 25.7 |
| Lags + $w$ + $v$ (4-week steps) | 0.006 | 0.170 | 0.119 | 1.42 | 0.95 | 91.8 |
| Lags + $w$ + $v$ (weekly steps) | -0.013 | 0.165 | 0.120 | 1.37 | 0.90 | 464.9 |
| *Mixed, slow $v$ ($\delta = 0.5$, $\phi = 0.97$ per week)* | | | | | | |
| Lags only | 1.650 | 0.367 | 0.107 | 3.42 | 0.00 | 0.1 |
| Lags + $w$ | 0.711 | 0.144 | 0.118 | 1.22 | 0.55 | 1.0 |
| Lags + $w$ + $v$ (8-week steps) | 0.503 | 0.194 | 0.121 | 1.60 | 0.85 | 22.9 |
| Lags + $w$ + $v$ (4-week steps) | 0.491 | 0.198 | 0.122 | 1.63 | 0.85 | 84.0 |
| Lags + $w$ + $v$ (weekly steps) | 0.479 | 0.203 | 0.122 | 1.66 | 0.85 | 468.4 |

TABLE S14

*S4 with asymmetric biome effects $\gamma = (0, 0.1, 0.2, 0.4)$ (30 replicates); otherwise as Table S5. Because the cell-weighted mean $\bar{\gamma}$ of $\gamma$ is not zero, the SHAP slope targets $(\bar{\gamma} + \gamma_b)/2$.*

| Estimator / biome | Mean est. | Cov. $\gamma_b$ | Cov. $\bar{\beta}_b$ | Detects sign |
|---|---|---|---|---|
| *Biome slopes, no latent effects* | | | | |
| biome 1 ($\gamma = +0.0$) | 0.013 | 0.53 | 0.90 | 0.00 |
| biome 2 ($\gamma = +0.1$) | 0.108 | 0.57 | 0.87 | 0.70 |
| biome 3 ($\gamma = +0.2$) | 0.197 | 0.57 | 0.63 | 0.93 |
| biome 4 ($\gamma = +0.4$) | 0.401 | 0.50 | 0.57 | 1.00 |
| *Biome slopes + $u$ + $w$* | | | | |
| biome 1 ($\gamma = +0.0$) | 0.012 | 0.67 | 0.87 | 0.00 |
| biome 2 ($\gamma = +0.1$) | 0.120 | 0.60 | 0.93 | 0.73 |
| biome 3 ($\gamma = +0.2$) | 0.213 | 0.57 | 0.73 | 0.97 |
| biome 4 ($\gamma = +0.4$) | 0.420 | 0.47 | 0.87 | 1.00 |
| *SVC: biome means + $\eta$ + $u$ + $w$* | | | | |
| biome 1 ($\gamma = +0.0$) | 0.013 | 0.87 | 1.00 | 0.00 |
| biome 2 ($\gamma = +0.1$) | 0.121 | 0.83 | 1.00 | 0.37 |
| biome 3 ($\gamma = +0.2$) | 0.209 | 0.77 | 1.00 | 0.77 |
| biome 4 ($\gamma = +0.4$) | 0.413 | 0.77 | 1.00 | 1.00 |
| *XGBoost SHAP slope (targets $(\bar{\gamma} + \gamma_b)/2$)* | | | | |
| biome 1 ($\gamma = +0.0$) | 0.111 | 0.30 | 0.30 | 0.00 |
| biome 2 ($\gamma = +0.1$) | 0.147 | 0.60 | 0.80 | 0.87 |
| biome 3 ($\gamma = +0.2$) | 0.186 | 0.67 | 0.67 | 0.97 |
| biome 4 ($\gamma = +0.4$) | 0.228 | 0.10 | 0.03 | 1.00 |
| *XGBoost SHAP slope, interactions credited* | | | | |
| biome 1 ($\gamma = +0.0$) | 0.052 | 0.80 | 0.83 | 0.00 |
| biome 2 ($\gamma = +0.1$) | 0.124 | 0.67 | 0.93 | 0.80 |
| biome 3 ($\gamma = +0.2$) | 0.205 | 0.63 | 0.80 | 0.93 |
| biome 4 ($\gamma = +0.4$) | 0.286 | 0.43 | 0.37 | 1.00 |
| *XGBoost median SHAP, raw temperature*[‡] | | | | |
| biome 1 ($\gamma = +0.0$) | -0.226 | 0.03 | 0.00 | 0.00 |
| biome 2 ($\gamma = +0.1$) | -0.174 | 0.00 | 0.00 | 0.00 |
| biome 3 ($\gamma = +0.2$) | -0.077 | 0.00 | 0.00 | 0.03 |
| biome 4 ($\gamma = +0.4$) | 0.092 | 0.00 | 0.00 | 0.67 |